Johannes U. Lange

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RESEARCH INTERESTS

Dark Energy, Gravitational Lensing, Galaxy Formation, Statistical Methods and Machine Learning

POSITIONS

American University 08/2024 – present
Assistant Professor
University of Michigan 09/2022 – 07/2024
Leinweber Center for Theoretical Physics Fellow
Stanford University 09/2021 – 08/2022
Stanford–Santa Cruz Cosmology Postdoctoral Fellow
University of California, Santa Cruz 09/2019 – 08/2021
Stanford–Santa Cruz Cosmology Postdoctoral Fellow

EDUCATION

Yale University 08/2014 – 08/2019
M.Sc., M.Phil, Ph.D. in Astronomy
Ruprecht-Karls-Universität Heidelberg 09/2012 – 08/2014
Master of Science in Physics
Freie Universität Berlin 10/2009 – 08/2012
Bachelor of Science in Physics

TEACHING

Instructor, American University Fall 2026
Course: Reality after Einstein
Instructor, American University Fall 2025
Course: Data Mining and Machine Learning for Natural Sciences
Instructor, American University Spring 2025, Spring 2026
Course: Changing Views of the Universe
Instructor, American University Fall 2024
Course: Modern Physics
Postdoctoral Course on STEM Teaching, University of Michigan Winter 2024
The Postdoctoral Short Course on College Teaching in STEM is a comprehensive 10-week program for postdocs to teach effectively as future faculty members.
Adjunct Lecturer, University of Michigan Fall 2023
Course: Naked-Eye Astronomy
Certificate of College Teaching Preparation, Yale University 2014-2019
The Certificate of College Teaching Preparation (CCTP) is an opportunity for graduate students to engage in a comprehensive training program in effective college teaching.
Teaching Fellow, Yale University Spring 2016, Spring 2018
Course: Astrostatistics and Data Mining

Teaching Fellow, Yale University Fall 2017, Fall 2014
Course: Introduction to Astronomical Observing
Teaching Fellow, Yale University Fall 2015
Course: Introduction to Cosmology
Teaching Fellow, Yale University Spring 2015
Course: Gravity, Astrophysics, and Cosmology

ADVISING

Elena Perez Alejandro (undergraduate) 2026 – present
Topic: The Physical Origins of Galaxy Assembly Bias
Patrick Connelly (undergraduate) 2026
Topic: Supermassive Black Hole Growth in IllustrisTNG
Kimiko Queen (undergraduate) 2026
Topic: The Lensing-is-Low Problem with DECADE
Simona Sotiri (undergraduate) 2025 – 2026
Topic: Testing Non-Linear Cosmology with FLAMINGO
Henry Gray (undergraduate) 2025 – 2026
Topic: The Dependence of Assembly Bias on Galaxy Properties
Alisun Coldiron (undergraduate) 2025
Topic: Galaxy-Halo Connection in FLAMINGO
Abigail Fisher (undergraduate) 2025 – 2026
Topic: Photometric Redshift Calibration
Alexandra Wells (undergraduate, co-adviser) 2024 – 2025
Topic: Cosmology from Non-Linear Scales
Alexandra Doytcheva (undergraduate) 2023 – 2024
Topic: Galaxy Clustering and Control Variates
Filomela Gerou (undergraduate) 2022 – 2024
Topic: Galaxy Clustering and Control Variates
Gilad Pifko (undergraduate) 2022 – 2023
Topic: Relationship between Galaxy and Dark Matter Halo Size
Simon Wu (undergraduate) 2022 – 2023
Topic: Gravitational Lensing Contribution from Subhalos
Garv Shah (undergraduate) 2022 – 2023
Topic: Boosting Importance Nested Sampling with Neural Networks
Juliana Karp (undergraduate) 2022 – 2023
Topic: Anisotropic Satellite Galaxy Quenching
Diana Blanco (graduate, co-adviser) 2021 – 2025
Topic: Photometric Redshift Calibration
Enia Xhakaj (graduate, co-adviser) 2019 – 2023
Topic: Gravitational Lensing

FIRST-AUTHOR PAPERS

[15] J. U. Lange et al. "A Unified Photometric Redshift Calibration for Weak Lensing Surveys Using the Dark Energy Spectroscopic Instrument". *ApJ* 998.2, 263 (Feb. 2026).

[14] J. U. Lange et al. "Cosmological Constraints from Full-Scale Clustering and Galaxy-Galaxy Lensing with DESI DR1". *arXiv e-prints*, arXiv:2512.15962 (Dec. 2025).

[13] J. U. Lange et al. "Systematic Effects in Galaxy-Galaxy Lensing with DESI". *The Open Journal of Astrophysics* 7, 57 (July 2024).

[12] J. U. Lange. "NAUTILUS: boosting Bayesian importance nested sampling with deep learning". *MNRAS* 525.2 (Oct. 2023), pp. 3181–3194.

[11] J. U. Lange et al. "Constraints on S_8 from a full-scale and full-shape analysis of redshift-space clustering and galaxy-galaxy lensing in BOSS". *MNRAS* 520.4 (Apr. 2023), pp. 5373–5393.

[10] J. U. Lange et al. "Five per cent measurements of the growth rate from simulation-based modelling of redshift-space clustering in BOSS LOWZ". *MNRAS* 509.2 (Jan. 2022), pp. 1779–1804.

[9] J. U. Lange et al. "On the halo-mass and radial scale dependence of the lensing is low effect". *MNRAS* 502.2 (Apr. 2021), pp. 2074–2086.

[8] J. U. Lange et al. "Cosmological Evidence Modelling: a new simulation-based approach to constrain cosmology on non-linear scales". *MNRAS* 490.2 (Dec. 2019), pp. 1870–1878.

[7] J. U. Lange et al. "New perspectives on the BOSS small-scale lensing discrepancy for the Planck ΛCDM cosmology". *MNRAS* 488.4 (Oct. 2019), pp. 5771–5787.

[6] J. U. Lange et al. "Updated results on the galaxy-halo connection from satellite kinematics in SDSS". *MNRAS* 487.3 (Aug. 2019), pp. 3112–3129.

[5] J. U. Lange et al. "Maturing satellite kinematics into a competitive probe of the galaxy-halo connection". *MNRAS* 482.4 (Feb. 2019), pp. 4824–4845.

[4] J. U. Lange et al. "Brightest galaxies as halo centre tracers in SDSS DR7". *MNRAS* 473.2 (Jan. 2018), pp. 2830–2851.

[3] J. U. Lange et al. "Evidence for Non-stellar Rest-frame Near-IR Emission Associated with Increased Star Formation in Galaxies at z ~1". *ApJL* 819.1, L4 (Mar. 2016).

[2] J. U. Lange and M.-C. Chu. "Can galactic dark matter substructure contribute to the cosmic gamma-ray anisotropy?" *MNRAS* 447.1 (Feb. 2015), pp. 939–947.

[1] J. Lange and M. Pohl. "The average GeV-band emission from gamma-ray bursts". *A&A* 551, A89 (Mar. 2013).

STUDENT-LED PAPERS

[3] K. Queen, A. V. Fisher, and J. U. Lange. "Revisiting the Mass and Scale Dependence of the "Lensing is Low" Effect with DECADE". *arXiv e-prints*, arXiv:2610.23828 (Oct. 2026).

[2] A. Doytcheva, F. V. Gerou, and J. U. Lange. "High-precision Galaxy Clustering Predictions from Small-volume Hydrodynamical Simulations via Control Variates". *ApJ* 977.2, 184 (Dec. 2024).

[1] J. S. M. Karp, J. U. Lange, and R. H. Wechsler. "Anisotropic Satellite Galaxy Quenching: A Unique Signature of Energetic Feedback by Supermassive Black Holes?" *ApJL* 949.1, L13 (May 2023).

OTHER CO-AUTHOR PAPERS

[35] D. Blanco et al. "The Power of DESI for Photometric Redshift Calibration: A Case Study with KiDS-1000". *The Open Journal of Astrophysics* 9 (Sept. 2026), p. 63965.

[34] A. Semenaite et al. "Joint cosmological fits to DESI-DR1 full-shape clustering and weak gravitational lensing in configuration space". *The Open Journal of Astrophysics* 9 (May 2026), p. 62334.

[33] DESI Collaboration et al. "Data Release 1 of the Dark Energy Spectroscopic Instrument". *AJ* 171.5, 285 (May 2026).

[32] A. Porredon et al. "DESI-DR1 3 × 2 -pt analysis: consistent cosmology across weak lensing surveys". *The Open Journal of Astrophysics* 9 (Apr. 2026), p. 61342.

[31] N. P. A. P. Emas et al. "Validation of the DESI-DR1 3x2-pt analysis: scale cut and shear ratio tests". *The Open Journal of Astrophysics* 9 (Mar. 2026), p. 59886.
[30] A. Polzin et al. "Picture an Astronomer: Best Practices for Retaining Talent in Astrophysics". *arXiv e-prints*, arXiv:2512.24465 (Dec. 2025).
[29] R. Ruggeri et al. "Clustering redshift distribution calibration of weak lensing surveys using the DESI-DR1 spectroscopic dataset". *arXiv e-prints*, arXiv:2512.15963 (Dec. 2025).
[28] K. Mitra et al. "BASILISK IV. No S_8 Tension with Satellite Kinematics". *arXiv e-prints*, arXiv:2512.14889 (Dec. 2025).
[27] S. J. Rauhut et al. "Testing gravitational physics by combining DESI DR1 and weak lensing datasets using the E_G estimator". *The Open Journal of Astrophysics* 8, 149 (Oct. 2025).
[26] S. Heydenreich et al. "Lensing Without Borders: Measurements of galaxy-galaxy lensing and projected galaxy clustering in DESI DR1". *arXiv e-prints*, arXiv:2506.21677 (June 2025).
[25] M. Kwiecien et al. "Improving galaxy cluster selection with the outskirt stellar mass of galaxies". *PRD* 111.12, 123524 (June 2025).
[24] C. Blake et al. "The DESI-Lensing Mock Challenge: large-scale cosmological analysis of 3x2-pt statistics". *The Open Journal of Astrophysics* 8, 24 (Mar. 2025).
[23] S. Chen et al. "Analysis of DESI×DES using the Lagrangian effective theory of LSS". *PRD* 110.10, 103518 (Nov. 2024).
[22] K. Mitra, F. C. van den Bosch, and J. U. Lange. "BASILISK II. Improved constraints on the galaxy-halo connection from satellite kinematics in SDSS". *MNRAS* 533.3 (Sept. 2024), pp. 3647–3675.
[21] S. Yuan et al. "Redshift evolution and covariances for joint lensing and clustering studies with DESI Y1". *MNRAS* 533.1 (Sept. 2024), pp. 589–607.
[20] Y. Wang et al. "Measuring the Conditional Luminosity and Stellar Mass Functions of Galaxies by Combining the Dark Energy Spectroscopic Instrument Legacy Imaging Surveys Data Release 9, Survey Validation 3, and Year 1 Data". *ApJ* 971.1, 119 (Aug. 2024).
[19] DESI Collaboration et al. "The Early Data Release of the Dark Energy Spectroscopic Instrument". *AJ* 168.2, 58 (Aug. 2024).
[18] E. Xhakaj et al. "Cluster cosmology without cluster finding". *MNRAS* 530.4 (June 2024), pp. 4203–4218.
[17] DESI Collaboration et al. "Validation of the Scientific Program for the Dark Energy Spectroscopic Instrument". *AJ* 167.2, 62 (Feb. 2024).
[16] B. Hadzhiyska et al. "Synthetic light-cone catalogues of modern redshift and weak lensing surveys waith ABACUSSUMMIT". *MNRAS* 525.3 (Nov. 2023), pp. 4367–4387.
[15] R. Ruggeri et al. "A data compression and optimal galaxy weights scheme for Dark Energy Spectroscopic Instrument and weak lensing data sets". *MNRAS* 525.3 (Nov. 2023), pp. 3865–3878.
[14] K. Wang et al. "Evidence of galaxy assembly bias in SDSS DR7 galaxy samples from count statistics". *MNRAS* 516.3 (Nov. 2022), pp. 4003–4024.
[13] DESI Collaboration et al. "Overview of the Instrumentation for the Dark Energy Spectroscopic Instrument". *AJ* 164.5, 207 (Nov. 2022).
[12] S. Huang et al. "The outer stellar mass of massive galaxies: a simple tracer of halo mass with scatter comparable to richness and reduced projection effects". *MNRAS* 515.4 (Oct. 2022), pp. 4722–4752.
[11] E. Xhakaj et al. "Beyond mass: detecting secondary halo properties with galaxy-galaxy lensing". *MNRAS* 514.2 (Aug. 2022), pp. 2876–2890.
[10] K. Dawson et al. "Snowmass2021 Cosmic Frontier White Paper: High Density Galaxy Clustering in the Regime of Cosmic Acceleration". *arXiv e-prints*, arXiv:2203.07291 (Mar. 2022).

[9] A. Leauthaud et al. "Lensing without borders - I. A blind comparison of the amplitude of galaxy-galaxy lensing between independent imaging surveys". *MNRAS* 510.4 (Mar. 2022), pp. 6150–6189.
[8] K. Wang et al. "Concentrations of dark haloes emerge from their merger histories". *MNRAS* 498.3 (Nov. 2020), pp. 4450–4464.
[7] F. C. van den Bosch, J. U. Lange, and A. R. Zentner. "Basilisk: Bayesian hierarchical inference of the galaxy-halo connection using satellite kinematics - I. Method and validation". *MNRAS* 488.4 (Oct. 2019), pp. 4984–5013.
[6] K. Wang et al. "How to optimally constrain galaxy assembly bias: supplement projected correlation functions with count-in-cells statistics". *MNRAS* 488.3 (Sept. 2019), pp. 3541–3567.
[5] A. R. Zentner et al. "Constraints on assembly bias from galaxy clustering". *MNRAS* 485.1 (May 2019), pp. 1196–1209.
[4] D. Campbell et al. "The galaxy clustering crisis in abundance matching". *MNRAS* 477.1 (June 2018), pp. 359–383.
[3] A. S. Villarreal et al. "The immitigable nature of assembly bias: the impact of halo definition on assembly bias". *MNRAS* 472.1 (Nov. 2017), pp. 1088–1105.
[2] E. J. Nelson et al. "Where Stars Form: Inside-out Growth and Coherent Star Formation from HST Hα Maps of 3200 Galaxies across the Main Sequence at $0.7 < z < 1.5$". *ApJ* 828.1, 27 (Sept. 2016).
[1] I. G. Momcheva et al. "The 3D-HST Survey: Hubble Space Telescope WFC3/G141 Grism Spectra, Redshifts, and Emission Line Measurements for ~100,000 Galaxies". *ApJS* 225.2, 27 (Aug. 2016).

INVITED TALKS

Particle Theory & Cosmology Seminar 04/2026
Johns Hopkins University
Astronomy Seminar 05/2025
Carnegie EPL
Cosmology Seminar 04/2025
University of California, Berkeley
CTC Seminar Series 03/2025
University of Maryland
Physics Colloquium 04/2024
University of Hawaii
ITP Cosmology Seminar 12/2023
Ruprecht-Karls-Universität Heidelberg
Frontiers of Nested Sampling Workshop 07/2023
42nd International Workshop on Bayesian Inference and Maximum Entropy Methods in Science and Engineering
Webinar Series 06/2023
National Observatory in Rio de Janeiro
Early Career Researcher Cosmology Seminar 11/2022
Korea Astronomy and Space Science Institute
HEAP Seminar 12/2021
University of Utah
Astronomy Colloquium 09/2021
Swinburne University of Technology

Growth of Structure Webinar 07/2021
University of California, Santa Cruz
Growth of Structure Webinar 06/2021
University of California, Santa Cruz
Research Progress Meeting 01/2019
Lawrence Berkeley National Laboratory
CCAPP Seminar 01/2019
Center for Cosmology and AstroParticle Physics
BCCP Seminar 09/2018
University of California, Berkeley
The Galaxy-Halo Connection Across Cosmic Time 07/2017
Kavli Institute for Theoretical Physics

OUTREACH

Public Observing Nights at AU, Washington, DC 2024 – present
KIPAC Public Lecture, Palo Alto, CA 07/2022
Class at Stanford Splash, Palo Alto, CA 11/2021
Talk at Astronomy on Tap, New Haven, CT 06/2019
Talk at Institute for Learning in Retirement, New Haven, CT 04/2019
Talks at Leitner Family Observatory, New Haven, CT 02/2018, 05/2019
Talks at Open Labs Science Cafe, Yale University, New Haven, CT 10/2017, 04/2019
Member of Open Labs, Yale University, New Haven, CT 2016 – 2019
Tutor at New Haven Reads, New Haven, CT 2015 – 2018
Member of UCSB Physics Circus, UC Santa Barbara, Santa Barbara, CA 2012

HONORS AND AWARDS

Scialog Fellow, Research Corporation For Science Advancement
Brouwer Ph.D. Thesis Prize, Yale University
Cosmology Fellowship, UC Santa Cruz and Stanford University
Graduate Fellowship Program, Kavli Institute for Theoretical Physics
Henry A. Smith Fellowship, Yale University
DAAD (German Academic Exchange Service) Scholarship
Deutschlandstipendium National Scholarship Program
Ernst Reuter Scholarship, Free University of Berlin

LEADERSHIP ROLES

Co-Chair of the DESI Sustainability Committee 2026 – present
Co-Chair of the Roman External Synergies Science Focus Group 2026 – present
Co-Chair of the NASA Cosmic Structure Science Interest Group 2025 – present
Co-Chair of the DESI desilike Topical Group 2024 – present
Co-Chair of the DESI C^3 Working Group 2022 – 2024

COMMITTEE MEMBERSHIP

Stanford Physics Equity & Inclusion Committee	2021 – 2022
DESI Professional Development Mentoring Program	2021 – present
DESI Early Career Scientists Committee	2021 – 2022
UCSC Astronomy Department Colloquium Committee	2019 – 2020
Yale Graduate Admissions Committee	2018 – 2019

REVIEWER SERVICE

Astronomy & Astrophysics
The Astrophysical Journal
Discover Space
Journal of Cosmology and Astroparticle Physics
Monthly Notices of the Royal Astronomical Society
The Open Journal of Astrophysics
Physical Review Letters
DESI Internal Reviewer
National Science Foundation

SCIENTIFIC ORGANIZING COMMITTEES

NASA's Physics of the Cosmos Early Career Workshop	09/2025
KICP Workshop	08/2023
Lensing at Different Scales: Strong, Weak, and Synergies Between the Two	
Michigan Cosmology Summer School 2023	06/2023
KITP Online Conference	08/2020
The Galaxy-Halo Connection Across Cosmic Time: Recent Updates	
KIPAC Online Workshop	07/2020
Precision Measurements and Modeling of Lensing and Clustering in the DESI Era	

REFERENCES

Frank C. van den Bosch
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Andrew P. Hearin
Argonne National Laboratory
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Alexie Leauthaud
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Dragan Huterer
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